

Data Analytics Fundamentals – Intermediate Level Question Paper

Maximum Marks: 100

Section A – Short Answer Questions (2 × 10 = 20 Marks)

Answer any ten questions. Each question carries 2 marks.

1. 1. What is the difference between a list and a NumPy array in Python?
2. 2. How do you read a CSV file into a Pandas DataFrame?
3. 3. Write a Python code to find the mean and standard deviation of a NumPy array.
4. 4. What is the purpose of the head() and tail() functions in Pandas?
5. 5. How can missing values be handled in a Pandas DataFrame?
6. 6. Explain the use of groupby() in Pandas with an example.
7. 7. What are the main components of a Matplotlib plot?
8. 8. Differentiate between plt.plot() and plt.scatter().
9. 9. What is a heatmap in Seaborn? Give one use case.
10. 10. Define correlation. How can you visualize it using Seaborn?

Section B – Medium Answer Questions (5 × 6 = 30 Marks)

Answer any five questions. Each question carries 6 marks.

11. 1. Write a Python program to create a NumPy array of shape (3,3) with random integers and find its transpose, mean, and sum.
12. 2. Explain different types of joins (inner, outer, left, right) in Pandas with an example.
13. 3. Plot a line graph using Matplotlib for x = [1,2,3,4,5] and y = [2,4,6,8,10] with titles, axis labels, and grid lines.
14. 4. Using Seaborn, visualize the distribution of a variable using both distplot and boxplot. Explain the difference in what they show.
15. 5. Explain the steps involved in performing Exploratory Data Analysis (EDA) on a dataset.
16. 6. Discuss how to detect and remove outliers using Pandas and Seaborn.

Section C – Long Answer / Coding Questions (5 × 10 = 50 Marks)

Answer any five questions. Each question carries 10 marks.

17. 1. Write a Python program using Pandas to: Load a dataset from CSV, display basic information and statistics, handle missing values, and group data by a categorical column and find mean of numerical columns.
18. 2. Create a NumPy array with 20 random numbers between 10 and 100. Find all even numbers, replace numbers greater than 80 with 80, and display sorted ascending and descending arrays.
19. 3. Using Matplotlib: Plot a bar chart showing sales of five products, add labels, title, and color customization, and save the plot as an image file.
20. 4. With Seaborn and an inbuilt dataset (like tips or iris): Create a pairplot and heatmap, and interpret what relationships can be observed from the plots.
21. 5. Consider a dataset sales.csv containing columns: Date, Product, Units_Sold, and Revenue. Read the dataset, find total revenue per product, plot a bar graph for visualization, and identify the product with maximum sales.

Section D – Data Analytics Case Study (1 × 10 = 10 Marks)

Case: You are provided with a dataset containing information about students' marks across subjects.

Columns: Name, Maths, Science, English, Total, Average, Grade.

Tasks:

22. 1. Load the dataset using Pandas.
23. 2. Compute Total and Average marks for each student.
24. 3. Assign grades based on average: A: ≥ 90 , B: 80–89, C: 70–79, D: < 70 .
25. 4. Visualize average marks using a bar chart and grades using a count plot.